

LAB TASK 1

Question 1:

```
#include<iostream>

#include<stdlib.h>

#include<ctime>

using namespace std;

class Inventory{
private:
    int size;

    // float avg;

    int* arr;
public:
    Inventory() {
        cout<<"Enter the size";

        cin >> size;

        // size = sz;

        arr = new int[size];

        for(int i = 0; i < size; i++) {
            // *(arr + i) = (rand() % 100) + 1;

            *(arr + i) = i+1;
        }
    }
}
```

```
float avg() {  
    int sum = 0;  
    for(int i = 0; i < size; i++) {  
        sum += *(arr + i);  
    }  
    return sum / size;  
}
```

```
int criticalProducts() {  
    int count = 0;  
    for(int i = 0; i < size; i++) {  
        if(*(arr + i) < avg()) count++;  
    }  
    return count++;  
}
```

```
int findTopSellingProduct() {  
    int topSell = *arr;  
    for(int i = 0 ; i < size; i++) {  
        if(*(arr + i) < topSell) topSell = *(arr + i);  
    }  
    return topSell;  
}
```

```
int findSecondBestSeller() {  
    int top = findTopSellingProduct();
```

```

int second = *arr;

for(int i = 0; i < size; i++) {
    if(* (arr + i) < second && *(arr+i) < top) second = *(arr+i);
}

return second;
}

```

```

void sortByPopularity() {
    for(int i = 0; i < size ; i++) {
        for(int j = 0; j < size; j++) {
            if(*(arr + i) > *(arr + j)) {
                int temp = *(arr+i);
                *(arr+i) = *(arr+j);
                *(arr+j) = temp;
            }
        }
    }

    cout<<"Sorted array is: ";
    for(int i = 0; i < size; i++) {
        cout<<*(arr + i)<<" ";
    }
}

```

```

~Inventory() {

```

```

        delete arr;
    }
};

int main() {
    srand(time(0));

    Inventory I1;

    cout<<"Average Stock level of inventory is: "<<I1.avg()<<endl;

    cout<<"Number of critical products are: "<<I1.criticalProducts()<<endl;

    cout<<"Top Selling product is: "<<I1.findTopSellingProduct()<<endl;

    cout<<"Second most top selling product is:" << I1.findSecondBestSeller()<<endl;

    I1.sortByPopularity();
}

```

Task 2:

```

#include<iostream>

using namespace std;

class Arr{
private:
    int size;

    int* arr;

public:

```

```
Arr(int sz) {  
    size = sz;  
    arr = new int[size];  
    cout<<"enter values ";  
    for(int i = 0; i < size; i++) cin >> arr[i];  
}
```

```
~Arr() {  
    delete arr;  
}
```

```
bool descending() {  
    int count = 0;  
    int* des = new int[size];  
    for(int i = 0; i < size; i++) {  
        des[i] = arr[i];  
    }  
    for(int i = 0; i < size ; i++) {  
        for(int j = 0; j < size; j++) {  
            if(*(des + i) > *(des + j)) {  
                int temp = *(des+i);  
                *(des+i) = *(des+j);  
                *(des+j) = temp;  
            }  
        }  
    }  
    for(int i = 0; i < size; i++) {
```

```

        if(arr[i] == des[i]) count++;
    }
    if(count == size) return true;
    else return false;
}

bool ascending() {
    int count = 0;

    int* aes = new int[size];

    for(int i = 0; i < size; i++) {
        aes[i] = arr[i];
    }

    for(int i = 0; i < size ; i++) {
        for(int j = 0; j < size; j++) {
            if(*(aes + i) < *(aes + j)) {
                int temp = *(aes+i);
                *(aes+i) = *(aes+j);
                *(aes+j) = temp;
            }
        }
    }

    for(int i = 0; i < size; i++) {
        if(arr[i] == aes[i]) count++;
    }

    if(count == size) return true;
    else return false;
}

```

```
};
```

```
int main() {
```

```
    Arr a1(5);
```

```
    if(a1.descending()) cout<<"it is in descending order"<<endl;
```

```
    else if(a1.ascending()) cout<<"It is in aescending order"<<endl;
```

```
    else cout<<"It is neither aescending nor descending"<<endl;
```

```
}
```